A Weekly Project Newsletter to Engage Episodic Participants in OSS Projects

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Abstract

Open Source Software (OSS) projects increasingly depend on a diverse set of contributors, including episodic participants who contribute intermittently. While dashboards and real-time communication tools support continuously active contributors, they often fail to serve the unique needs of episodic participants, who may struggle to remain informed and re-engage with project activity after periods of absence. In this paper, we examine the effect of a weekly, AI-generated, email-based newsletter intervention designed to improve awareness and engagement among episodic OSS contributors. We ground our approach in self-determination theory, emphasizing autonomy, competence, and relatedness as key motivational constructs. To inform the design of the Weekly Project Newsletter, we conducted a literature review of techniques that support communication in OSS projects, followed by a think-aloud study with OSS contributors, leading to a refined newsletter that summarizes recent activity in issues, pull requests, and commits, and highlights project milestones and community interactions. We evaluate the final design of our Weekly Project Newsletter through a three-week diary study involving nine episodic contributors. Our findings show that the newsletter improves participant awareness and perceived project connection. However, its influence on OSS contribution behavior remains limited without actionable prompts or surfacing of beginner-friendly tasks.

Keywords: Open Source Software, Episodic Contributors, Project

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1. Introduction

Open Source Software (OSS) is a societal good that has significantly accelerated digital advancement [1]. As a result, OSS is pervasive, forming the foundation of a wide span of critical technologies. It is in society's interest to foster a strong and growing OSS ecosystem by ensuring key projects are maintained and new projects are successfully created. The challenge is that OSS projects often rely on the work of volunteers who contribute their time in order to improve their own software development skills and to be part of a community [2, 3]. McDonald and Goggins note that the growth and participation of these community volunteers are the primary measure of OSS project health [4]. However, the turnover rate among OSS participants is high [5], while OSS projects often fail when they lose contributors, even if they are still useful to the community [6].

Researchers have devoted considerable effort to identifying factors that influence volunteer participation in OSS projects [7, 8, 9, 10]. The needs of project newcomers differ from those of project maintainers and highly active participants. Newcomers are more likely to join OSS projects that offer clear onboarding support, responsive maintainers, and recognition for their contributions [11, 12, 13, 14]. In contrast, long-term, highly active participants and project maintainers are influenced by burnout, project governance structures, and sustained community engagement [15, 6]. A separate category consists of episodic contributors (or participants), developers who engage with an OSS project intermittently, contributing in short bursts separated by extended periods of inactivity, often returning opportunistically in response to specific needs or interests [16, 17]. Episodic contributors require clear communication with project maintainers and flexibility in contribution timelines [18, 19].

Despite ongoing research, many OSS projects still struggle to consistently operationalize these insights in day-to-day community practices. Moreover, existing tools and practical interventions largely target project maintainers or first-time contributors, leaving other types of participants less directly supported [20, 11, 21].

In this paper, we introduce Weekly Project Newsletter, an AI-generated newsletter designed to keep periodic contributors engaged by summarizing key project updates and delivering them via email. Unlike existing tools that

focus on maintainers or newcomers, Weekly Project Newsletter targets contributors who participate intermittently, ensuring they stay informed without needing daily involvement. Each issue of the newsletter highlights important pull requests, discussions, and project milestones, making it easier for episodic contributors to remain engaged and contribute effectively.

We developed Weekly Project Newsletter using a systematic approach. First, we reviewed the literature to identify information relevant to OSS participants. We then created an initial version of the newsletter and conducted a think-aloud study with 9 OSS contributors, using their feedback to refine its content and format. After implementing these improvements, we ran a three-week diary study with 9 episodic contributors (different from the think-aloud study) to evaluate how Weekly Project Newsletter impacted their engagement and participation.

In summary, this paper makes the following set of contributions:

- Practical intervention to improve the retention and engagement of episodic contributors in OSS projects.
- Design of the Weekly Project Newsletter, a newsletter that keeps periodic contributors engaged by providing key project updates to help them stay informed and participate more effectively.
- Design insights derived from formative feedback with OSS contributors, clarifying which project signals in the Weekly Project Newsletter support awareness and perceived connection.
- Empirical evidence from a three-week diary study showing that the Weekly Project Newsletter improves awareness and perceived connectedness among episodic contributors, while also revealing limitations in encouraging increased project contribution.

Our intervention is grounded in Self-Determination Theory (SDT) [22], which emphasizes autonomy, competence, and relatedness as key drivers of sustained motivation. Weekly Project Newsletter supports these psychological needs by offering curated updates (competence), personalized and voluntary engagement (autonomy), and recognition of contributions (relatedness).

We begin the paper by describing the design of the Weekly Project Newsletter in Section 2, which includes three phases: an initial design, a think-aloud study, and a final design. In Section 3, we present our evaluation of the final Weekly Project Newsletter, followed by a discussion in

Section 4. In Section 5, we outline the threats to validity, and in Section 6, we describe related work. Finally, Section 7 provides the conclusion and outlines future work for the project.

The Weekly Project Newsletter is designed to operate over any standard GitHub repository artifacts, including issues, pull requests, commits, and releases, and does not assume project-specific conventions or customization. While our evaluation instantiates the newsletter on a set of OSS projects, the design itself is intended to be applicable to any GitHub-hosted project. A replication package that includes the code and scripts to create the Weekly Project Newsletter is available at: `https://github.com/vcu-swim-lab/weekly-project-news`

2. Weekly Project Newsletter Design

The purpose of the Weekly Project Newsletter is to provide open source contributors with relevant updates about their project repositories in a structured, digestible format. By delivering curated summaries, the newsletter is designed to help reduce perceived information overload and support continued connection to the project, especially for episodic contributors returning after a period of inactivity. This approach builds on prior work for disseminating project information to open source developers, such as dashboards [20, 23, 24], by offering a push-style format through email, which delivers updates directly to developers in a more accessible and actionable way.

Recently, e-mail newsletters on platforms like Substack have become popular across a variety of domains. Similarly, AI News is a newsletter that publishes a daily summary of high-frequency AI-related news from platforms like X and Reddit [25]. However, this daily frequency is often too frequent for the typical OSS episodic contributor. Prior research on mailing list engagement suggests that weekly updates effectively maintain subscriber interest without causing fatigue [26]. Daily updates, by contrast, can overwhelm contributors, especially those involved in multiple projects. Additionally, OSS repositories vary in activity levels, with some receiving only a few meaningful updates per week. A weekly schedule ensures that contributors receive a well-curated summary of relevant changes rather than sparse or redundant daily updates.

To sign up for the newsletter, a developer enters their email address and reference to the *public* GitHub repository they are interested in on the

Weekly Project Newsletter landing Web page: `https://buttondown.com/weekly-project-news`.

2.1. Phase 1: Initial Design

For the initial design, we first reviewed literature on communication strategies for OSS participants, focusing on tools that helped maintainers and contributors stay informed. To identify relevant prior work, we searched Google Scholar, the ACM Digital Library, and IEEE Xplore using keyword combinations such as "open source" paired with "dashboard," "contributor awareness," "project health," and "communication tools." We focused on papers that described concrete tools or techniques for communicating project information to OSS participants. We supplemented this keyword-based search with backward and forward snowballing from key references, in particular from Climate Coach [20] and related dashboard studies. This process was not intended as a systematic or exhaustive literature review, but as a targeted scan to inform the design of the newsletter. For instance, Guizani et al. developed a dashboard for tracking newcomer activity [21], while Qiu et al. analyzed GitHub transparency signals, such as stars and contributor activity, that influenced contributors' decisions to join a GitHub project [27]. More specifically, we drew direct inspiration from Climate Coach, a dashboard designed to help OSS participants monitor collaboration and maintain civil discourse [20]. Several of its features, such as tracking active and stale issues, analyzing conversational tone, and monitoring contributor activity, proved highly informative to OSS participants. However, adapting these concepts to a newsletter format required a different approach.

The initial design of the newsletter was split into four main sections: Issues, Pull Requests, Commits, and Contributors, with subsections of each described in Figure 1. The Issues section summarized recent activity, including newly opened issues, frequently discussed issues, long-standing issues, and recently closed issues. It also included an Issue Discussion Insights subsection that analyzed conversations for toxicity, assigning scores from 0 to 1. A score above 0.5 was considered significant and flagged for review, a threshold inspired by the CRAFT tool for conversation derailment [28]. The Pull Requests section provided summaries of open and closed pull requests and analyzed discussions for potential toxicity. The Commits section organized related commits from the past week based on commit messages. Finally, the Contributors section ranked authors based on their combined activity across

Initial Newsletter Structure

1. Issues
1.1 Open Issues – *summaries of groups of related issues opened in the last week*
1.2 Top 5 Active Issues – *summaries of the most commented issues*
1.3 Top 5 Quiet Issues – *summaries of the issues opened the longest within the last year*
1.4 Closed Issues – *summaries of the groups of related issues closed this week*
1.5 Issue Discussion Insights – *summaries of predicted toxic issue conversations in the last week*

2. Pull Requests
2.1 Open Pull Requests – *summaries of groups of related pull requests opened in the last week*
2.2 Closed Pull Requests – *summaries of groups of related pull requests closed in the last week*
2.3 Pull Request Discussion Insights – *summaries of predicted toxic discussions in pull request comments*

3 Commits – *summaries of groups of related commits from the last week based on commit messages*

4 Contributors – *rankings of contributors based on issues, pull requests, and commits, sorted by super-score*

Figure 1: Initial content of the Weekly Project Newsletter.

issues, pull requests, and commits, assigning a super-score that determined their placement in the rankings.

The decision to track Open Issues, Closed Issues, Open Pull Requests, and Closed Pull Requests was influenced by research on signals that attract contributors to open source projects [27]. Similarly, the idea of ranking the Top 5 Active Issues and Top 5 Quiet Issues was inspired by Climate Coach [20], which demonstrated how surfacing key discussions enhances engagement. The design of the Commits section was informed by the study of micro-collaborative coding practices in the OpenStack ecosystem [29], which provided insights into how multiple contributors can collaborate on the same commit. This motivated our design of the Weekly Project Newsletter to track and highlight such collaborative efforts in the form of commit contributions. Additionally, the sections for Issue Discussion Insights and Pull Request Discussion Insights were inspired by research on stress and burnout in open source communities [30], which emphasized the importance of detecting toxic interactions in GitHub issue discussions and their potential impact on developer well-being. By incorporating such an analysis,the Weekly Project Newsletter aimed to highlight discussions that could be harmful or stressful, helping to maintain a healthier open source environment. The Contributors section itself was inspired by research on how to attract and retain contributors through maintainers' dashboards, particularly the importance of understanding and tracking contributor activities and contributions [21]. By structuring updates around activity, responsiveness, and interaction quality, the Weekly Project Newsletter aimed to help contributors stay informed and engaged without overwhelming them with information.

Weekly Project Newsletter's design also follows principles from SDT [22], as a formative design lens, to address the motivational needs of episodic contributors. SDT is a theory of human motivation that explains why people initiate, sustain, or disengage from activities based their basic psychological needs. SDT argues that motivation is more durable when three core needs are supported: *competence*, the feeling of being effective and capable; *autonomy*, the experience of choice and volition; and *relatedness*, the sense of connection with others. Prior work shows that system design can influence how these needs are supported in technology-mediated settings [31]. When these needs are not supported, motivation often weakens or becomes externally driven. In particular, we designed the Weekly Project Newsletter to align with SDT as follows:

- *Competence:* By summarizing pull requests, issues, and commits, the newsletter is intended to help contributors understand ongoing work, boosting their perceived competence.
- *Autonomy:* Subscription-based delivery and flexible engagement enable contributors to stay informed on their own terms.
- *Relatedness:* The inclusion of contributor recognition and community discussions is intended to fosters a sense of social belonging and connectedness to the OSS community.

The implementation consisted of several steps. First, we collected data using the GitHub REST API and stored it in a database. Next, we queried, sorted, and cleaned the data into a concise JSON format. This JSON was processed by ChatGPT (gpt-4o), which applied grouping and summarization prompts based on the content, generating the newsletter in Markdown format. We chose ChatGPT because it was (1) easy to use, (2) relatively inexpensive at the time (as newer GPT versions had been released), and (3) effective for our summarization needs. We used the standard OpenAI API and a temperature of 0, as we wanted consistent and repeatable outputs. The final Markdown file was sent to subscribers via the Buttondown API. To automate this process, a Linux server ran a CRON job every Monday early morning, handling data collection, cleaning, and newsletter distribution. We also relied on Buttondown to manage subscribing and unsubscribing to the newsletter.

2.2. Phase 2: Think-Aloud Study

After the initial design and implementation of the Weekly Project Newsletter, we conducted a think-aloud study to evaluate the initial design of the Weekly Project Newsletter by observing how OSS contributors engaged with a version generated from recent activity in a repository they had contributed to. Our goal was to assess the clarity, relevance, and usefulness of the content. Participants reviewed the newsletter while verbalizing their thoughts, allowing us to identify which elements provided value, which required revision or removal, and what additional content might improve the newsletter's utility.

2.2.1. Think-Aloud Study Methodology

Throughout this paper, we use the term "episodic contributor" to refer to a participant whose contribution history to a given project is limited in volume and who also self-reports infrequent engagement, defined as contributing less than a few times per month. Both criteria are used together in Section 3 to identify episodic contributors for the diary study. The think-aloud study described here did not restrict recruitment to episodic contributors. We instead recruited a broader population of OSS contributors in order to evaluate the initial newsletter design. The insight that the newsletter is most useful for episodic contributors emerged from this study and motivated the participant criteria used in Section 3. We selected study participants that are contributors to popular open source repositories on GitHub. We used a script that extracted email addresses from commits made during the past week in repositories with at least 500 stars on GitHub. We also required each repository to have at least 50 commits during that week. The star threshold was intended to target popular, widely used projects from which we could recruit contributors, and the commit threshold was intended to exclude non–actively developed projects such as practice problem collections and toy projects. Active projects were preferred because they were more likely to have a sufficient pool of recent contributors for recruitment, and because active projects are the setting in which the newsletter is most likely to be useful in practice. We then cross-referenced these email addresses with the repository documentation and author sections to ensure accuracy and personalize our outreach. The recruitment email provided details about the study, including its purpose, duration, compensation, and voluntary nature. It also specified that participation required a 30-minute recorded Zoom session. The number of invitations we sent was approximately 300–400, though the exact number was not logged. Interested contributors were asked to complete a pre-study survey. The pre-study survey screened candidates based on four criteria: experience as active maintainers, involvement in open source projects with varying community activity, willingness to provide verbal feedback, and availability for the study sessions. From the 47 received responses, we selected 9 participants based on their project activity and demographic diversity to ensure a range of perspectives. We contacted them to schedule their Zoom interviews. The participants reflected a diverse set of OSS projects: Llama.cpp, node.js, OpenXLA, PyTorch, Copilotkit, and Shadcn-ui.

All think-aloud study participants were male and represented a wide range

Characteristic	**Count**
Gender	
Male	9
Age Range	
18-24	2
25-34	3
35-44	2
45-54	1
55-64	1
Project Tenure	
0–6 months	2
6–12 months	3
1–2 years	1
2–3 years	1
>5 years	2
Contribution Frequency	
Daily	3
Several times a week	3
Once a week	1
A few times a month	2
Less frequently	–
Developer Experience	
1–3 years	2
7–10 years	1
>10 years	6
Open Source Experience	
<1 year	2
1–3 years	2
4–6 years	1
7–10 years	2
>10 years	2

Table 1: Demographics of think-aloud study participants ($N = 9$).

of ages, project tenure, contribution frequency, and both general software development and open source experience. Table 1 summarizes the participant demographics. The Think-Aloud study consisted of a 30–60 minute Zoom call, during which a scripted set of questions was read to participants to gather their responses while they interacted with the Weekly Project Newsletter constructed for their respective OSS project of interest. First, participants were introduced to the study process and provided verbal consent to be recorded with both audio and video. This study was approved by the Virginia Commonwealth University Institutional Review Board (IRB) under protocol HM20030697. They were then asked an initial set of questions about their experience as open source developers. Each participant described their repository, their role in it, and how they stayed up to date with its activity.

Participants discussed the most important aspects of open source repositories that influenced their contributions, answering questions like: 1) To provide some background on your repository, what is *[repository]* and how do you contribute to it? 2) How do you currently stay up to date with repository activity? Do you review all issues and commits? and 3) What are the most important components of an open source GitHub repository when contributing, e.g., pull requests, issues, or other elements?

2.2.2. Think-Aloud Study Results

Participants described their current methods for staying updated with repository activity, often emphasizing the challenges of managing information flow. One participant noted, *"Yeah, so that's how I currently do it, because CoPilot Kit is a fairly new project, I think. So it doesn't have that many issues or pull requests, so that it's fairly easy to keep track of what's happening. But it's growing very fast. So every day I can see a lot of new issues and new pull requests coming in. So currently I just manually go through all this whenever I get time and I try to contribute."* This highlights how, as repositories scale, maintaining awareness of relevant updates becomes increasingly difficult. This is a challenge the newsletter aims to address by consolidating information efficiently.

An episodic contributor discussed their approach for staying up-to-date with an OSS project, stating *"Normally I will look through the issues. Yes, I rarely get down to the commit level unless I'm looking for something very specific. And I'll be honest, I haven't contributed much. As a matter of fact, I've contributed exactly one PR to this, and it was a documentation change.*

I'm very much an opportunistic open source contributor." This perspective demonstrates how episodic contributors often rely on issue tracking to determine where they can contribute without staying deeply attentive to project activity.

The main part of the session involved participants receiving a copy of Weekly Project Newsletter generated for their specific repository, which we identified through the pre-survey. They were then instructed to verbalize their thoughts while reading it. Finally, after reviewing the newsletter, each participant answered a series of follow-up questions to capture their concise feedback. They were asked to provide clearer opinions on the newsletter, including their likes and dislikes, as well as their perceptions of the effectiveness of specific sections. The following questions were asked:

1. Which sections of the newsletter did you find most important?
2. What would you like to see added to the newsletter that was not present?
3. What would you like to see removed from or clarified in the newsletter?
4. Did the newsletter provide a good sense of the project's progress and overall well-being?
5. Do you think a weekly newsletter, sent every Monday at 8 a.m., is appropriate? Or would you find it more useful if sent daily, monthly, biweekly, etc.?

One of the participants emphasized the importance of recognizing non-code-based contributions within the newsletter, stating *"Yeah, I think I mentioned the like, non-code-based contributions, so valuable review and review comments. Pull request comments and issue comments and commit comments, possibly like just any, anything that's not noise, but like that's a contribution to the project that includes people who triage issues and stuff. Right. Like in theory that person would never again get noticed otherwise, but they may be providing a lot of value to a project."* (TA-P4) This feedback showed that while this design iteration of the Weekly Project Newsletter effectively communicated progress in terms of code contributions, it did not fully capture the broader collaborative efforts that sustain an OSS project. By enhancing the "Contributors" section to include issue and pull request comments within the last week, the newsletter could improve relatedness and better recognize

contributors, which in turn will provide a more comprehensive view of the project's health and community well-being.

Another participant highlighted the need for a dedicated section on version releases, stating, *"Well, like I said, the first thing I'd want to see is releases, which is missing entirely. So especially any major version ones. Right. So I would say releases should be top with and maybe could just inline the release notes."* (TA-P4) This feedback indicated that tracking version updates is a priority for some OSS developers who wanted a clearer view of a project's evolution over time, which corresponds to a sense of project health. Additionally, the participant identified closed pull requests as one of the most important existing sections as well as merged pull requests as a potential new section, explaining, *"And then after that, merged pull requests and then closed ones."* (TA-P3) Their perspective suggests that restructuring the newsletter to prioritize releases and merged pull requests could make it more informative and better aligned with the needs of open source contributors.

After conducting interviews with all participants, we transcribed the sessions verbatim. Three of the authors then independently engaged in an axial coding process [32], using memoization to support the development of an emergent coding scheme. Each coder generated an initial set of categories based on open coding. These initial categories were subsequently reviewed and synthesized during a collaborative, in-person discussion to resolve discrepancies and refine the coding framework. Through this iterative process, we identified five primary thematic categories: formatting, content, clarity, engagement, and community health. Within each category, we defined subcategories and applied focused coding to the interview data. A summary of the thematic analysis from the think-aloud study is presented in Table 2.

2.3. Phase 3: Final Design

The think-aloud study revealed that the newsletter's primary value lies in supporting episodic contributors, i.e., developers who engage intermittently with open source projects and benefit from low-overhead summaries of project activity. Unlike newcomers and maintainers, who typically prefer pull-based communication for onboarding or day-to-day coordination, episodic contributors benefit more from push-based communication that helps them stay informed [33, 18]. Guided by this insight, we revised the newsletter to improve its clarity, relevance, and usability, with particular attention to surfacing project health signals, enhancing navigability, and recognizing

Category / Subcategory	# Devs / # Mentions	Example Quote
Clarity	8 / 59	*"Also, you may want a releases section because one of the items mentioned in the commits section is version 22.50 features like knowing, being able to highlight as its own section, like project releases, because those may or may not come from pull requests."* (TA-P4)
Feature Suggestions	8 / 28	
Readability and Understanding	7 / 27	
Section Calculation / Creation	3 / 4	
Community Health	4 / 5	*"But if there's no baseline for me to compare against, then I can't tell if the project is doing better than normal or about average... I mean, it's just giving me a point in space. But until I can see trends, I can't know."* (TA-P1)
Sense of Project Progress	3 / 3	
Toxicity	2 / 2	
Content	9 / 54	*"So here it actually summarized them and grouped them into a particular grouping that I kind of like as a, like, as a value add rather than just listing the individual pull requests."* (TA-P2); *"Yeah, I can see how this is useful to summarize all the issues rather than go through them. Saves time, especially if you're a regular contributor."* (TA-P3)
Clustering	1 / 2	
Issues, PRs, Commits, Contributors	9 / 31	
Usefulness of Information	4 / 4	
Summaries	8 / 17	
Engagement	7 / 9	*"That, to me, doesn't feel like it has a ton of value unless, like, because I'm not a core maintainer, even if I, you know, do code review, my approval is not necessarily going to make it [...]"* (TA-P8)
Contributor Roles	6 / 6	
Interactivity	3 / 3	
Formatting	9 / 26	*"Typically when people are scanning through documents, they kind of like spend the most time at the top. They kind of also skim in an F shaped pattern. So that's kind of burned a little bit of how much I want to be spending time on this as a newsletter."* (TA-P8); *"[...] it would be nice to see, like a table of contents before I scroll down."* (TA-P6)
Buttondown	4 / 6	
Headings	3 / 6	
Scheduling	7 / 7	
Structure	4 / 7	

Table 2: Summary of the results of the think-aloud study.

contributor activity. The changes we made aimed to better support the SDT needs of autonomy, competence, and relatedness.

Participants consistently requested clearer indicators of project momentum, such as recent version releases and trends in issue resolution. In response, we added two new sections: *Recent Version Releases*, which highlights newly tagged releases with links to changelogs, and *Other Noteworthy Updates*, a flexible section for surfacing project-specific events or milestones not captured by GitHub metadata. These additions aim to help readers assess the project's state at a glance, which also support contributors' perceived SDT need of competence.

To reduce information overload, we shortened several sections and revised how data is presented. The *Contributors* section was reduced to a ranked list of the top ten contributors from the past month. Rankings are computed using a *super-score*, a sum of the number of recent commits, pull requests, issues, and comments authored. This ensures recognition of both code and non-code contributions while improving brevity. Recognizing diverse forms of participation, especially non-code efforts, promotes SDT relatedness by validating the social and collaborative aspects of contribution. Similarly, open and closed pull requests are now divided into two groups: three *key pull requests*, ranked by number of associated commits, and a summarized list of the remainder. Each key pull request includes its merge status and links to associated commits. Commit details, which were previously listed separately, are now embedded within pull request summaries to avoid redundancy.

We also redefined issue categories to reflect more temporally relevant activity. *Top 5 Active Issues* now refers to those with the most comments during the past week, rather than overall comment count. *Top 5 Stale Issues* are now defined as open issues with no updates in the last 30 days, replacing the previous metric based on issue age alone. These refinements make the issue sections more useful for identifying timely engagement opportunities. This temporal relevance reinforces SDT autonomy by helping contributors selectively engage with current and actionable discussions.

Formatting improvements were made to address navigational concerns. We introduced an interactive table of contents, allowing readers to jump to specific sections within the email. Section headers were updated to better reflect standard terminology, for example, "quiet issues" was renamed to "stale issues." Date formatting was standardized, and each newsletter now includes a link to the previous edition to support continuity.

In response to concerns about the limited scope of toxicity detection,

we expanded the analysis to include both open and closed issues and pull requests. This adjustment better captures harmful discussions that may have been closed or locked due to moderation. By helping contributors avoid stressful or hostile interactions, this change also supports psychological safety, indirectly reinforcing SDT relatedness.

Finally, we addressed several formatting and generation issues caused by large inputs to the language model. For instance, prior versions occasionally included hallucinated links or ambiguous references such as "this issue." These problems were mitigated through prompt engineering and tighter control over input size. Link formatting was also corrected to ensure all references are valid and clearly displayed.

Figure 2 shows a screenshot of the final Weekly Project Newsletter as delivered to a subscriber. The final Weekly Project Newsletter design is significantly more concise than the initial iteration. The newsletter begins with a table of contents, followed by news section, including recent version releases and version information. It then proceeds to issue analysis, highlighting both the top five active issues, based on the number of comments during the current week, and the top five stale issues, defined as those with no updates in the past month. Next, issues opened and closed during the week are grouped and summarized, with accompanying discussion insights that analyze the tone of related comments and conversations. The newsletter then shifts focus to pull requests. Both open and closed pull requests are divided into two groups: three key pull requests, which are selected based on the number of associated commits, and a summarized list of the remaining pull requests. Each key pull request includes its merge status, a direct link to the pull request on GitHub, and a list of associated commits with clickable links to the diffs. Remaining pull requests are grouped and summarized in the same manner as issues. The pull request discussion insights section mirrors the issue discussion insights but is applied to both open and closed pull requests. Finally, the contributors section lists the top ten contributors from the past month, ranked by the number of commits, pull requests opened, issues opened, and comments authored. In combination, these changes enhance the SDT-aligned design by nurturing competence (through concise summaries), autonomy (through self-directed navigation), and relatedness (through contributor recognition).

I. News

1.1 Recent Version Releases:

The current version of this repository is v2.6.0

1.2 Version Information:

The PyTorch 2.6 release, created on January 29, 2025, introduces

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II. Issues

2.1 Top 5 Active Issues:

We consider active issues to be issues that that have been commented on most frequently within the last week. Bot comments are omitted.

1. **[compile] Modularize very long compilation**: This issue

● ● ●

2.2 Top 5 Stale Issues:

We consider stale issues to be issues that has had no activity within the last 30 days. The team should work together to get these issues resolved and closed as soon as possible.

1. **DISABLED**

● ● ●

2.3 Open Issues

This section lists, groups, and then summarizes issues that were created within the last week in the repository.

Issues Opened This Week: 96

Summarized Issues:

- **API and Export Issues in PyTorch:** The need for a robust

● ● ●

2.4 Closed Issues

This section lists, groups, and then summarizes issues that were closed within the last week in the repository. This section also links the associated pull requests if applicable.

Issues Closed This Week: 57

Summarized Issues:

- **Test Failures and Disabling Tests**: This topic covers

● ● ●

2.5 Issue Discussion Insights

This section will analyze the tone and sentiment of discussions within this project's open and closed issues that occurred within the past week. It aims to identify potentially heated exchanges and to maintain a constructive project environment.

1. **`Torch.distributed.tensor.parallel.Style.Rowwiseparallel` when send tensor, the use of the underlying operator is `aten.to.dtype_layout`, lack of distributed strategy. But after adding a policy, `AsyncCollectiveTensor` no placements error**
   - Toxicity Score: 0.55 (Frustration over unresolved issues,

III. Pull Requests

3.1 Open Pull Requests

This section provides a summary of pull requests that were opened in the repository over the past week. The top three pull requests with the highest number of commits are highlighted as 'key' pull requests. All other pull requests are grouped based on similar characteristics for easier analysis.

Pull Requests Opened This Week: 163

Key Open Pull Requests

1. [test] 2: This pull request, titled "[test] 2," aims to address and

● ● ●

Other Open Pull Requests

- **MXFP8 and MXFP4 Support in PyTorch**: This topic

● ● ●

3.2 Closed Pull Requests

This section provides a summary of pull requests that were closed in the repository over the past week. The top three pull requests with the highest number of commits are highlighted as 'key' pull requests. All other pull requests are grouped based on similar characteristics for easier analysis.

Pull Requests Closed This Week: 241

Key Closed Pull Requests

1. Fix SEGFAULT when None arg was passed in

● ● ●

Other Closed Pull Requests

- **Manylinux 2.28 and GCC 13 Update**: This pull request

● ● ●

3.3 Pull Request Discussion Insights

This section will analyze the tone and sentiment of discussions within this project's open and closed pull requests that occurred within the past week. It aims to identify potentially heated exchanges and to maintain a constructive project environment.

1. **export method**
   - Toxicity Score: 0.55 (Defensive responses, critique of

● ● ●

IV. Contributors

4.1 Contributors

Active Contributors:

We consider an active contributor in this project to be any contributor who has made at least 1 commit, opened at least 1 issue, created at least 1 pull request, or made more than 2 comments in the last month.

If there are more than 10 active contributors, the list is truncated to the top 10 based on contribution metrics for better clarity.

Contributor	Commits	Pull Requests	Issues	Comments
malfet	197	61	2	216

Figure 2: Screenshot of the Weekly Project Newsletter for PyTorch 2.6.0, edited to omit nonessential details for brevity.

3. Evaluation

After completing the final design of the Weekly Project Newsletter, we aimed to evaluate its effectiveness in the context of episodic OSS contributors. Specifically, we conducted a diary study to address the following research questions:

1. RQ1: How do episodic contributors perceive the usefulness and relevance of the Weekly Project Newsletter in keeping them informed about project updates?

2. RQ2: Does the Weekly Project Newsletter contribute to sustained participation and awareness among episodic contributors?

Characteristic	**Count**
Gender	
Male	9
Age Range	
18–24	2
25–34	3
35–44	2
45–54	2
Project Tenure	
0–6 months	3
6–12 months	1
1–2 years	1
2–3 years	2
3–5 years	1
>5 years	1
Contribution Frequency	
Daily	–
Several times a week	–
Once a week	–
A few times a month	–
Less frequently	9
Developer Experience	
1–3 years	1
4–6 years	2
7–10 years	1
>10 years	5
Open Source Experience	
<1 year	2
1–3 years	2
4–6 years	2
>10 years	3

Table 3: Demographics of diary study participants ($N = 9$).

3.1. Participants

We recruited using direct email a total of 9 episodic contributors. We recruited participants from well-known open source projects, selected based on the number of stars and level of activity on GitHub: Llama.cpp, Matplotlib, VSCode, Kubernetes, node.js, PyTorch, Terraso-mobile-client, and Tensorflow. We targeted well known and active projects for two reasons. First, the newsletter depends on meaningful weekly activity such as issues, pull requests, and releases in order to produce useful content. Projects with little activity would yield sparse newsletters and would not allow participants to evaluate the artifact in a realistic way. Second, episodic contributors are primarily found in projects that are large enough to attract occasional participation. The number of stars and the level of activity therefore served as proxies for the presence of this target population. These episodic contributors were distinct from the participants in the think-aloud study described in Section 2.2. We chose participants that had contributed to a repository within the past year, but were within the bottom half of active contributors in terms of number of contributions in the last year. We additionally required participants to self-report infrequent engagement, defined as contributing less than a few times per month. In combination with their contribution history, we used this self-report to classify each participant as episodic according to the definition introduced in Section 2.2. Table 3 summarizes the demographics of the diary study participants. Prior to the start of the study, participants responded to demographic questions, including how long they had been involved with their GitHub project, how frequently they worked on it, their years of experience as developers, and their experience in open source development. The results are that the participants came from diverse OSS and industry backgrounds. Their project involvement ranged from under 6 months to over 5 years. While most had significant development experience (1 to over 10 years), their open source experience was generally limited, with several reporting less than 3 years of contribution. Most reported infrequent engagement with their projects.

3.2. Procedure

This study was conducted over a three-week period to examine how regular, targeted communication in the form of Weekly Project Newsletter affects contributor awareness and engagement in OSS projects. At the beginning of the study, each participant identified a GitHub repository aligned with their interests or current involvement. This ensured that each newsletter they

Category	**Likert Question**	**Open-Ended Question**
Usefulness	How useful was this week's newsletter in keeping you informed about project updates?	What did you find most useful and least useful in this week's newsletter regarding project updates?
Relevance	How relevant was this week's newsletter in keeping you informed about GitHub project topics?	What did you find are the most and least relevant parts of this week's newsletter in keeping you informed about project updates?
Raised Awareness	Has this week's newsletter helped you stay aware of changes, issues, or upcoming tasks in the project?	What aspects of the newsletter do you find most useful for improving your awareness about the project?
Influenced Contribution	Has this week's newsletter influenced your decision to contribute to the project in the near future?	What would make this week's newsletter more helpful in encouraging your contribution to the GitHub project?
Suggested Changes	N/A	If you could change one thing about this week's newsletter, what would it be?
General Helpfulness/Additional Comments	N/A	Did you find the newsletter helpful overall? Also add any additional comments here.

Table 4: Survey questions grouped by evaluation category.

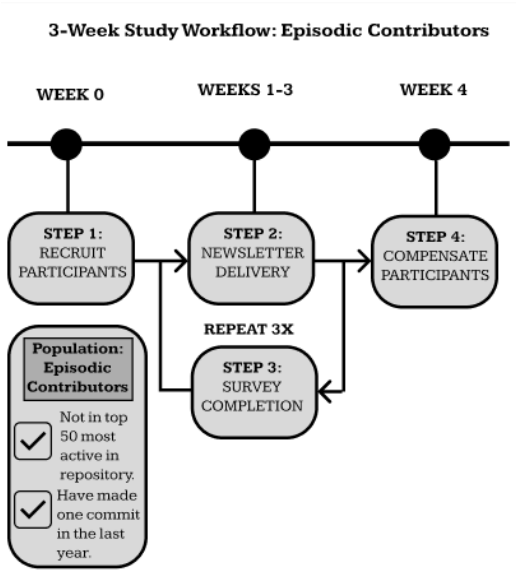

Figure 3: Overview of the diary study procedure.

received was contextually relevant, increasing the likelihood of meaningful contribution. Each Monday, for up to three weeks, participants received a custom newsletter summarizing recent activity in their selected repository via email. The structure of Weekly Project Newsletter followed the final design described in the previous section.

After reviewing the newsletter, participants completed a brief survey consisting of both Likert scale and open-ended questions, as shown in Table 4. The Likert questions assessed perceptions of the newsletter's usefulness, relevance, and impact. The open-ended questions allowed participants to elaborate on which aspects of the newsletter were most or least helpful, suggest improvements, and propose features that might increase their participation. Participants were compensated based on their level of participation: those who completed all three weeks received a $50 gift card, those who completed two weeks received $20, and those who completed one week received $10.

Participants reviewed the newsletter and completed the survey on a weekly basis. We analyzed the Likert scale data from the most recent week each participant completed. Qualitative responses were drawn from all three weeks to capture a broader range of perspectives. The study process is outlined in Figure 3. While nine participants initially enrolled, only seven completed all

three weeks of the survey. The remaining two participants completed only one week.

3.3. Findings

3.3.1. Useful Sections and Signals

Participants consistently identified the Active Issues section as the most useful part of the newsletter. They found the timeliness of the section particularly helpful for staying aware of issues that required immediate attention in the repository. One participant noted, *"The most useful part was definitely the issues section, specifically the 'Top 5 Stale Issues' and 'Open Issues' subsections. These help me stay informed about relevant, unresolved problems where I can contribute."* This consistent preference suggests that OSS contributors value brief, actionable, and time-sensitive information that directly aligns with their motivation to help.

The Closed Pull Requests section emerged as the second most commonly mentioned useful feature, especially for contributors seeking a clear snapshot of recent changes in the project. One participant remarked, *"The most relevant updates are 'Closed Issues' and 'Closed Pull Requests' since they show what is changing in the project."* Another echoed this sentiment, stating, *"The 'Closed Issues' and 'Closed Pull Requests' sections are great for tracking changes in the project each week."*

3.3.2. Perceived Usefulness and Relevance (RQ1)

Most participants found the newsletter useful and relevant for staying up to date with their open source project, reflected in average usefulness and relevance ratings of 3.89 and 3.56, respectively. Participants appreciated the summaries and likened the newsletter to a stand-up meeting in its ability to surface key updates.

However, some found sections like contributor statistics less relevant to their needs. A few also preferred interactive formats over static content, suggesting that while the newsletter works well for many, personalization could improve its effectiveness across diverse contributor roles, e.g., by focusing only on a particular part of the project that is relevant to a specific episodic participant.

3.3.3. Impact on Participation and Awareness (RQ2)

Participants overwhelmingly indicated that the newsletter increased their awareness of project activity, which aligns with the category's highest average

score of 4.22. Many responses emphasized that the regular updates helped them stay connected to the project's trajectory, even when they were not actively contributing. Some mentioned that the newsletter offered a clear view of what was changing week to week, providing a sense of continuity. One participant summarized this effect by saying *"Sure all information is already publicly available, but having a summarization about the key issues, open issues, stale issues, etc. is very useful."*

However, this heightened awareness did not translate into a strong influence on contribution, which received the lowest average score of 2.67. Many participants cited time constraints as a barrier to engaging further, even when they found the content informative. Others felt the newsletter lacked clear calls to action. Several respondents suggested surfacing unassigned tasks, highlighting beginner-friendly issues, or explicitly identifying what needs to be worked on. The disconnect between awareness and actual contribution reflects a common challenge in OSS, being informed does not always translate into action. Factors like timing, confidence in making a meaningful contribution, or the lack of clear next steps may still hold contributors back, even when they feel up to date.

Table 5 shows a summary of participant feedback across four key evaluation categories: usefulness, relevance, raised awareness, and influence on contribution. Each average score represents the aggregated Likert responses across all survey participants, while the quotes offer a qualitative snapshot that captures the tone and insights behind the numeric ratings.

3.3.4. Qualitative Insights

While participants gave numerical ratings on aspects such as perceived usefulness and relevance, their open-ended responses uncovered additional insights that brought nuance to the study. These responses reinforce and contextualize the findings for RQ1 and RQ2, particularly around how participants perceived the newsletter's value and how it may have shaped their awareness and participation.

Theme 1: Information Overload. A frequently mentioned suggestion was that the newsletter felt too long, particularly when displaying large sections of issues or pull requests. One participant noted, *"A little 'Show More' button could be added to some of the issues and pull request subsections that display too many items. Right now, some of these sections make the newsletter feel overly long."* This perception of information directly related to RQ1; it

Category	Avg. Score	Positive Sentiment	Negative Sentiment
Usefulness	3.89	*"It is good in giving a quick overview about what is currently happening in the codebase."*	*"really long newsletter. In this case, I have the newsletter related to kubernetes, which is a very popular and active github repo. This makes it that each section is filled with a huge amount of information."*
Relevance	3.56	*"Very insightful as it contains everything someone would want to read about a repo."*	*"Least relevant was the toxic conversations section, since there were none flagged."*
Raised Awareness	4.22	*"It is almost like a digital version of an agile stand-up. You get an overview of the key issues and what other people are working on in a relatively small timeframe."*	*"please include a title or summary title for each pull requests as that would allow people to be more easily informed about the updates on the things they are interested by picking the pull request to read by the PR's (summary) title."*
Influence on Contribution	2.67	*"Definitely, the number of pull requests closed this week. Seeing such a large number makes me feel like this project has a lot of momentum and people actively contributing. It also reassures me that if I tried to contribute, I wouldn't be ignored."*	*"This week's newsletter would be more helpful in encouraging my contribution if it included a list of beginner-friendly issues that new contributors could work on."*

Table 5: Average participant scores (on a 5-point Likert scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree) and representative positive and negative sentiments by evaluation category.

impacted how participants judged the newsletter's usefulness and relevance. Even so, participants still found substantial value in the content itself.

Theme 2: Desire for Beginner-Friendly/Actionable Content. Participants frequently asked for beginner-friendly issue suggestions. While open issues were included, many were not accessible to newcomers. One participant shared, *"If I could change one thing about this week's newsletter, it would be to add a list of issues suitable for beginners to contribute to."* This suggests that episodic or first-time contributors may benefit from clearer guidance and targeted calls to action as compared to project managers who might prefer an overview of all activity. This quote was also selected as the representative for the Influence on Contribution category. This explains the gap between the newsletter increasing awareness and its inability to motivate contribution. While participants became more informed, they lacked clear pathways to engage.

Theme 3: Interest in Project Health and Trends. Several participants expressed a strong interest in understanding how a repository evolves over time, emphasizing that a newsletter could do more than just present a static snapshot of current activity. Instead, they hoped it would surface broader patterns and trends that reflect the health and direction of the project. For instance, one participant suggested, *"You could also show that there were an X [percent] more issues closed/opened in comparison with last week,"* highlighting a desire for more dynamic comparative metrics. This type of trend-focused information could help contributors quickly assess momentum or stagnation in the project's workflow and adjust their involvement accordingly.

These comments suggest that while the current design supports passive consumption of updates, some users desire a more analytical view of the project. This is consistent with a pattern in which contributors with limited time prefer formats that help them make informed decisions quickly. Adding comparative metrics and trends could further support this decision-making flow and appeal to a broader set of contributor needs.

Theme 4: Format and Delivery Preferences. While much of the participant feedback focused on content, several participants emphasized the importance of how that content was delivered. Although most participants appreciated the concise format and actionable updates, some participants still had improvements in mind. Participants expressed a desire for more alternative

newsletters formats like mobile-friendly interfaces, web applications, or even the ability to create the newsletters whenever needed rather than have them be created on a set day. This indicates that when the content is valuable, rigid formats or interface limitations can hinder engagement. As one participant noted, *"If I don't look at this email for the next couple of days it will definitely already be outdated. I have only once a week an overview of the current status,"* highlighting the limitations of fixed interval updates for those with more time-sensitive and immediate needs. The suggestions showcase that even when the information is valuable, limitations in its accessibility can negatively impact how people engage with it.

Overall, these observations suggest that the newsletter is a strong source of relevant and timely information, especially in raising awareness of repository activity. Participants generally found it useful, but several noted that its current structure can feel inflexible. For episodic contributors in particular, this rigidity may make it harder to engage meaningfully, especially when beginner friendly content or flexible delivery options are lacking. Improving adaptability by adding features like dynamic issue filtering without assignees or on-demand generation could help the newsletter better serve this audience.

4. Discussion

In this section, we discuss how our findings compare to those of similar techniques, and present implications for OSS projects and for episodic OSS contributors.

4.1. Comparison with Existing Approaches

Our findings reveal important distinctions between the Weekly Project Newsletter and existing tools like dashboards and project health monitors. Unlike dashboard approaches, such as Climate Coach [20], that require active checking and interpretation of metrics, the newsletter delivers concise summaries directly to contributors through a familiar medium. This aligns with prior research showing that episodic contributors prefer lightweight engagement mechanisms over tools requiring constant attention [18].

The newsletter format also addresses a fundamental challenge faced by episodic contributors: maintaining awareness with minimal time investment. While dashboards can provide extensive metrics, they typically target project maintainers and require users to actively seek out information. Research has shown that dashboards' effectiveness is limited by contributors' willingness to

actively check and interpret them [34], and they may even generate negative messaging among less active developers [35]. In contrast, the Weekly Project Newsletter proactively delivers relevant information to contributors who lack the time or motivation to regularly check project status through traditional channels. As one participant noted, it functions as "almost like a digital version of an agile stand-up," providing updates in a condensed time frame.

Similarly, real-time chat platforms like Slack and Discord, while widely adopted in OSS communities [36], can lead to information overload and fragmented discussions [37], which is challenging for episodic contributors who engage intermittently. The newsletter mitigates this by operating on a weekly basis with periodic summaries, helping episodic contributors avoid the "fire hose" effect of real-time communications while still staying informed of project issues.

4.2. Implications for Episodic Contributors

The findings suggest that Weekly Project Newsletter helps lower the barrier for episodic contributors by reducing cognitive overhead and surfacing key issues in a digestible format. Rather than requiring contributors to actively monitor dashboards or navigate GitHub threads, the newsletter provides curated updates that make it easier to understand where the project stands and what needs attention.

This form of regular, structured communication may also support re-engagement after periods of inactivity. By maintaining a sense of connection and visibility into project activity, the newsletter helps episodic contributors regain context without the burden of catching up manually [38]. However, the results also indicate that awareness alone is not always enough to motivate contribution. While participants felt more informed, many did not act on that awareness. This suggests that additional support, such as clearer next steps, encouragement, or task matching may be needed to translate passive engagement into active participation [39].

Framing the newsletter through the lens of SDT clarifies its value: it addresses motivational barriers to OSS participation by supporting contributors' needs for competence (by surfacing relevant information), relatedness (by recognizing contributions), and autonomy (by enabling opt-in, asynchronous engagement). Future work could strengthen these components further—for example, offering personalized contribution suggestions would better support autonomy and competence simultaneously.

4.3. Implications for OSS Project Maintenance

For maintainers, curated newsletters offer a lightweight and low effort community management strategy. Rather than relying solely on dashboards or scattered GitHub notifications, newsletters provide a centralized summary that keeps contributors informed and engaged. By highlighting recent activity, surfacing neglected issues, and reducing ambiguity about project priorities, newsletters can help maintain momentum and improve the visibility of ongoing work. This can be especially valuable when trying to attract contributors toward areas that need attention. This approach is particularly effective in large or loosely connected OSS communities where centralized coordination is limited. In such environments, consistent and accessible communication tools like newsletters can help create a shared understanding of project status, even among contributors with sporadic involvement.

Our findings also have implications for platform providers such as GitHub. Although GitHub offers notifications, release notes, and activity feeds, it does not currently provide a structured, periodic summary tailored to contributors who engage intermittently. A platform-integrated solution leveraging LLM summarization, could deliver repository-level weekly summaries to followers or prior contributors. Our results suggest that such summaries could improve awareness and perceived connection, particularly for episodic contributors.

5. Limitations

Several limitations may affect the interpretation of our findings. We group them below by standard validity categories.

Construct validity. This threat concerns whether our design features accurately reflect the concepts they intend to represent. We interpret episodic contribution and project health using surface metrics (e.g., recent activity, changelogs, comment counts), which may not capture deeper contributor intent or engagement. We mitigated this by grounding our design in prior work and participant feedback. We also aligned features with SDT, using its constructs, autonomy, competence, and relatedness, as a guiding framework.

Internal validity. Internal validity refers to factors within the study that could bias results. Our think-aloud study included only 9 participants, which may limit robustness. To mitigate this, we sampled developers with varied experience. Subjectivity in interpreting qualitative feedback is another risk, though we used thematic coding and consensus review. Additionally, although we addressed known language model errors, unobserved issues in

newsletter generation may remain. Our recruitment pipeline was not instrumented to record the exact number of candidate projects that met the inclusion criteria or the exact number of invitations sent, so these counts can only be reported as estimates. Replication therefore must rely on the documented recruitment procedure rather than on exact figures.

External validity. Our findings are based on a limited number of GitHub-hosted projects and contributors familiar with that ecosystem. As a result, the design may not generalize to other platforms or contributor types (e.g., first-time or non-code contributors). We also do not measure long-term behavioral impacts. Broader deployment and longitudinal studies are needed to assess generalizability.

6. Related Work

6.1. OSS Project Health and Episodic Contributors

The concept of project health in OSS contexts encompasses multiple dimensions, including technical quality, community vitality, and organizational sustainability [6, 40, 41]. In recent years, the health of the OSS project has gained the attention of researchers and practitioners [42, 42]. There are various factors that can affect the health of an OSS project, including the presence of a supporting community, dynamic roles, and clear communication [43, 44, 20]. These factors are interrelated and collectively contribute to the long-term success of OSS projects [45, 44, 46]. For example, community vitality can be seen in the active participation of users, developers, and other stake holders, which directly impacts the project's health, growth, and sustainability [47]. Despite this, the community and volunteer management of OSS projects is often divided and incomplete [48, 49, 50]. One key factor influencing OSS project health is communication among participants [51, 43, 52, 53]. Effective communication strategies help maintain engagement, resolve conflicts, and facilitate collaboration, all of which contribute to the sustainability of OSS projects [54].

To keep an open source community growing, there should be a heavy focus on the contributors [55]. Attracting newcomers and maintaining newcomers are the two most successful ways to foster open source project growth [24]. Newcomers who fix at least one issue in their first month have more than double the odds of becoming a Long Term Contributor [56]. In addition to attracting newcomers, ensuring that they do not take prolonged breaks is a necessary strategy since only around 21-26% of contributors will return to

an open source community if their break lasts for a year or more [56]. This abandonment of open source communities by episodic contributors leads to damaging effects for the project. [57]

6.2. Communication Approaches for OSS Participants

Effective communication is essential for the success of OSS projects, as contributors are often geographically dispersed and have many different roles [46, 53]. Clear and efficient communication fosters knowledge sharing, reduces misunderstandings, and improves project coordination [52]. To support this, dashboards can provide real-time insights into contributor activity, expertise, and project status, helping to improve transparency and collaboration [24].

However, while dashboards offer valuable information, their effectiveness depends on contributors actively checking and interpreting them [34]. Without a structured way to highlight key updates, important insights can be overlooked [35]. This limits their impact on participation and decision making. In addition, dashboards also cause more negative messages, especially among developers with low activity [35]. This suggests that performance boosts may come with unintended negative effects on developer well-being and community health, which can lead to an unwillingness to participate in OSS [58].

Real-time chat platforms like Slack, Discord, and IRC are widely adopted in OSS projects to facilitate quick, informal communication [21]. The tools support immediate feedback and fast problem-solving, which can be particularly helpful during active development phases or when urgent issues arise [59]. Additionally, these tools help build a stronger sense of community and belonging by enabling casual conversation which is essential in OSS projects [53]. However, they can also lead to information overload or fragmented discussions if not properly organized, especially in large projects with many contributors like OSS projects [37].

This work complements prior research on OSS dashboards, real-time communication tools, and onboarding support for OSS newcomers by focusing on episodic contributors and examining a push-based, periodic communication artifact rather than continuous monitoring or real-time interaction. Our results position AI-generated newsletters as a alternative lightweight mechanism for improving awareness and perceived connection with OSS projects.

7. Conclusions and Future Work

In this paper, we presented the Weekly Project Newsletter as a lightweight communication intervention aimed at supporting episodic contributors in open source software projects. Through an iterative design process and a three-week diary study, we showed that the newsletter improves contributor awareness and perceived connection to project activity. Participants valued concise summaries of issues and pull requests that reduced the effort required to stay informed after periods of inactivity. At the same time, our results indicate that increased awareness alone is not sufficient to consistently motivate contribution.

This study opens several opportunities for future research. One promising direction is the development of customized newsletters tailored to different user needs such as versions configured by maintainers, adapted to contributor history, or segmented by role (e.g., newcomers, reviewers, or consumers). Personalized content could make the communication even more relevant. Additionally, longitudinal studies are needed to examine whether repeated exposure to newsletters can foster or sustain contribution behavior over time. While this study captured short-term perceptions and effects, future work could explore how ongoing engagement with newsletters influences contributor trajectories, retention, and quality of participation in OSS projects. We could also assess the effects of a AI-generated newsletter compared to other baseline approaches (e.g., dashboards).

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